Zakaria Amgrout

AI Engineer · Data Scientist — LLM & Machine Learning Systems

Tangier, Morocco | +212 602 337 751 | amgrout.zakaria@gmail.com
linkedin.com/in/amgrout-zakaria | github.com/blueskullMRX | blueskullmrx.github.io/portfolio

Profile

AI Engineer with a Master's in Artificial Intelligence and Data Science (2026) and hands-on experience across the applied-AI stack: data pipelines (Kafka, Spark), deep learning (PyTorch, TensorFlow), retrieval-augmented generation and LLM orchestration, and containerised, monitored deployment. Track record spans healthcare AI, real-time big-data analytics, and LLM-driven automation, taking systems from prototype to production. Seeking an AI Engineer or Data Scientist role building ML/LLM systems.

Education

Master's Degree in Artificial Intelligence and Data Science Sep 2024 – Jul 2026
Faculty of Science and Technology, Tangier

Bachelor's Degree in Software Development Engineering Sep 2023 – Jul 2024
Faculty of Science and Technology, Tangier

Professional Experience

AI Engineer Intern **Feb 2026 – Jul 2026**
NTT DATA — Final-Year Master's Project (Hybrid) *Morocco*

- Built an end-to-end platform automatically generating JUnit test suites for Java/Spring Boot applications, combining deterministic static analysis with Retrieval-Augmented Generation to cut manual test-authoring effort
- Achieved a 97–98% average validation score on generated test suites
- Engineered the RAG pipeline: method-level analysis documents embedded (OpenAI text-embedding-3-small), indexed in Qdrant and retrieved via metadata-filtered semantic search, grounding LLM prompts in real project structure
- Designed a rule-based test-strategy selector over an AST-derived project model (JavaParser) covering type resolution, dependency injection and control-flow analysis
- Deployed as a distributed service (FastAPI, Celery/Redis, PostgreSQL, React/TypeScript, Docker) with monitoring via Prometheus and Grafana

AI Engineer **Jul 2025 – Aug 2025**
DIGIUP *Rabat, Morocco*

- Developed SEHATOKA, an AI-driven healthcare platform that improved patient diagnosis accuracy by 30% and cut workflow processing time by 40%
- Designed and deployed ML models for symptom-based patient routing and disease-risk prediction, and integrated LLMs into an NLP doctor chatbot for clinical decision support and automated patient management

Web Developer **Apr 2024 – Jun 2024**
Smart Automation Technologies Pvt. Ltd. *Tangier, Morocco*

- Built the resource-management module of a MEAN-stack platform for auditing hardware and logistics inventory

Projects

Enterprise Knowledge Assistant for Microsoft Teams (RAG) Team project, 2026

- Co-built a RAG-based AI assistant for Microsoft Teams that answers employee questions from company documents (SharePoint, PDF, Word, Excel) using Azure OpenAI, LangChain and Qdrant
- Enabled secure, permission-aware access so users only get answers from documents they are allowed to view; backend built with FastAPI, Celery and Redis

Real-Time Amazon Review Sentiment Analysis May 2025

- Built a real-time Kafka/Spark Streaming/MongoDB pipeline for review sentiment classification with Dockerised models

Facial Expression Detection Mar 2025 – Apr 2025

- Trained a CNN (TensorFlow/Keras) for emotion recognition; deployed as a real-time Streamlit webcam app

Technical Skills

LLMs & Generative AI: RAG, LangChain, OpenAI API, Hugging Face Transformers, prompt engineering, embeddings, semantic search, hybrid search, reranking, Azure OpenAI, Qdrant & vector databases

Machine / Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, SciPy, OpenCV, NLTK

Cloud & Automation: Microsoft Azure, AWS, Databricks, n8n

Data & Databases: Apache Kafka, Spark Streaming, ETL pipelines, Power BI, PostgreSQL, MongoDB, MySQL

Backend & MLOps: FastAPI, Flask, .NET Core, React, Celery, Redis, Microsoft Graph API, Bot Framework, Docker, Git, GitHub Actions, Linux, Prometheus, Grafana

Programming: Python, Java, C#, C++, JavaScript/TypeScript, SQL

Languages: Arabic (Native), English (Professional working proficiency), French (Professional working proficiency)